



Competition of Emergency Rapid Deflation Device- (ERDD-2023)

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Design Report Submitted By

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Nomenclature

CAD Computer Aided Design

ERDD Emergency Rapid Deflation Device

kgf Kilogram force

LTA Lighter than Air

RPM Rotations per minute

1 Mission Profile

The aerostat is attached to a tether at 0.2kgf buoyancy. On cutting the tether, the ERDD is activated. The airship will rise to a certain height till the system in place kicks in. Then it will stay at a particular height and then descend to the ground. Our aim is to deflate the aerostat as soon as possible without damaging the aerostat or payload.

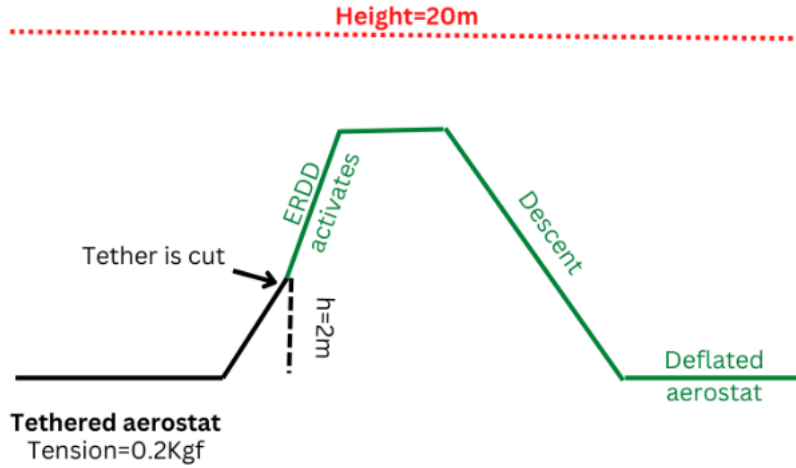


Figure 1: Mission Profile

2 Concept

The aerostat constitutes two below drawn ERDD mechanisms, one cylinder is attached at the top of the aerostat to suck helium out and a second cylinder is attached at the bottom of the aerostat to fill air in it so it gets heavier and touches the ground sooner. The motor rotates to unscrew a telescopic mechanism which pushes the propeller ahead. As the propeller moves ahead, it pops the mylar from the other end of the cylinder.

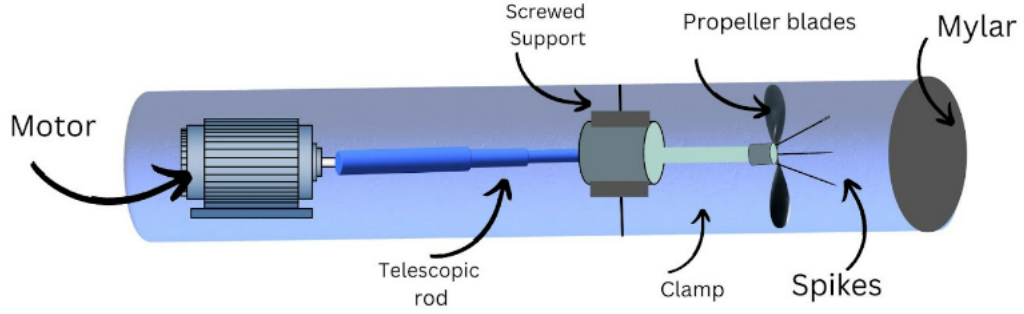


Figure 2: Diagram

2.1 Systems

2.1.1 Sensing System

Here we are using an ultrasonic sensor to measure changes in altitude or height upon tether breakage. By measuring the time it takes for an ultrasonic sound wave to travel from the sensor to a target surface below and back again, one can calculate the distance between the sensor and the surface. When the tether is cut, the sensor mounted on an aerostat, uses the change in distance measurements over time to calculate the object's change in altitude. It then sends a signal to the Arduino with the help of established connections which in turn activates the ERDD.

2.1.2 Processing System

As soon as the tether is cut, the change in voltage connected to the sensor completes the circuit and a signal is sent to the Arduino UNO board. We have placed an Arduino UNO board for processing in case of momentary tension loss in the tether due to transients, or atmospheric gusts. To take care of this, the controller is pre-programmed to wait for a specific amount of time; and triggers the circuit only if the tether tension is not regained during this time. This delay has been kept as programmable so that it can be used in various conditions and the delay can be set accordingly. Once the relay gets switched on after the pre-programmed time elapses, it completes the third and final circuit.

2.1.3 Actuation system

Our preliminary design comprises a hollow cylindrical tube that is installed into the aerostat. The aerostat is sealed by attaching a layer of mylar at one

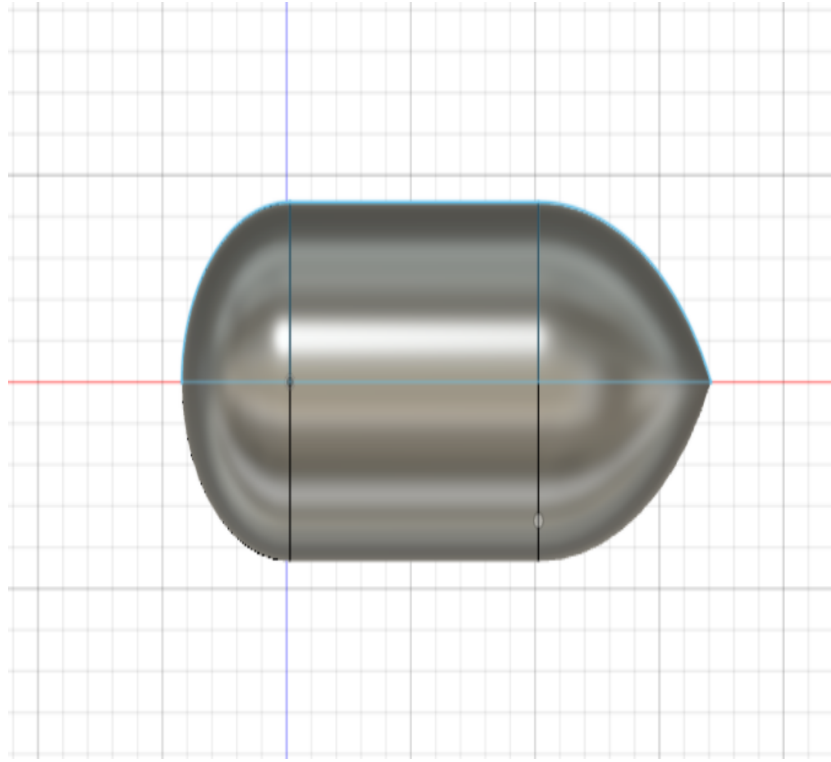


Figure 3: Shape of Aerostat

end of the cylindrical tube whilst the ERDD comprises a motor attached to a rod with telescopic mechanism which is further attached to a fan. Once the motor starts operating on receiving a signal from arduino UNO, the rod with the telescopic mechanism will extend. The telescopic rod is connected to a simple and small cylindrical structure with grooves engraved on it and is supported by clamps. With every rotation of the motor, the grooves will unscrew and the rod will extend further. The fan at the end of the rod, has spikes embedded in front of it. The fan will be propelled forward into the mylar patch and the spikes will draw a tear into the patch. After the patch is removed, the fan will suck the helium out of the aerostat at its maximum rpm.

2.2 Tabular Sequence

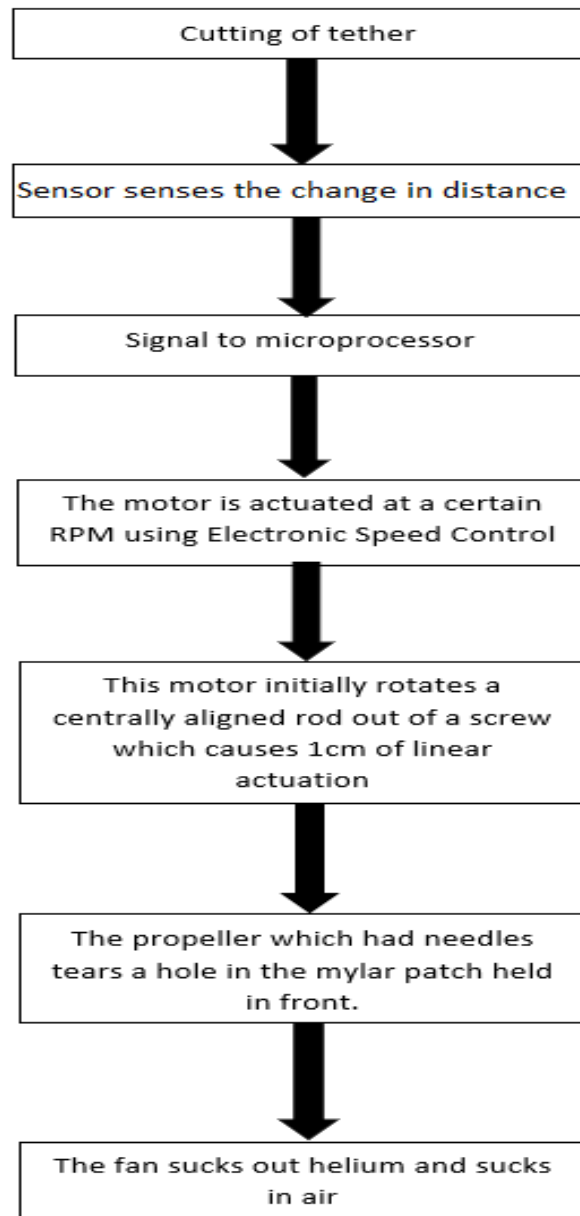


Figure 4: Tabular Sequence

2.3 Electronic Components and Connections

In this system, once the ultrasonic sensor detects the distance of 1.3 meters from the ground, it will send an input signal to arduino. Arduino will then send an output signal to start the working of motors at that very instant. The arduino is connected to L298N motor driver first which will provide enough voltage for both motors to rotate the the required rpm. Once the aerostat deflates and comes down at a distance of 1.3 meters from ground again, the system will shut down itself.

Here in this github repository is the arduino code for this setup:

https://github.com/Kanwalpreet0312/ERDD/blob/main/ERDD_code.ino

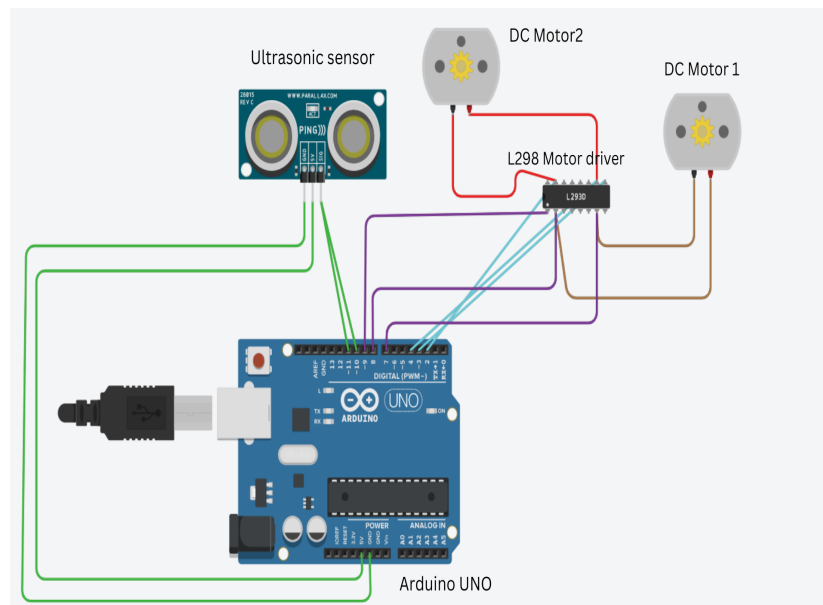


Figure 5: Design placement in aerostat

2.4 Lift Calculations

Input	Max Size	Rectangular envelope	4*2	m
	Gondola mass	m_g	50	g
	Fin mass	m_f	0	g
	Avionics	m_w	152.5	g
	System mass	m_s	472	g
	Payload	m_p	5	g
	Ballast	m_B	0	g
	Fabric weight	m_fab	0.1890956	g
	miscellaneous	m_m	200	g
	Heaviness mass	m_H	0	g
	Air density	rho_air	1.293	kg/m ³
	Lifting gas density	rho_He	0.18	kg/m ³
	Fabric density(m	rho_fab	0.03401	kg/m ²
	Gravity	g	9.81	m/s ²
Shape Definition	Parameter	Variable name	Value	Units
	Volume	Venv	1.01	m ³
	Surface Area	SA	5.56	m ²

Figure 6: Weight and Shape Estimation

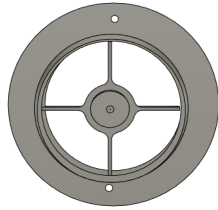
Calculations	Parameter	Variable Name	Value	Units
	Effective Density	rho_air-rho_he	1.113	kg/m ³
	Volume	Venv	1.01	m ³
	Raw Lift	RL	11.02771	N
	Mass of all systems	m_tot	0.880982	kg
	Weight of all systems	W_tot	8.642434	N
	Total lift		2.385280	N

Figure 7: Shape of Aerostat

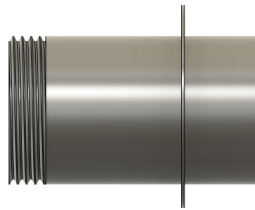
2.5 CAD diagram for concept visualization

Components:

1. Hollow cap of the cylinder: The top of the cylinder is engraved with external grooves which are to be attached to the internal grooves of the cap. A patch of mylar is to be placed between the hollow section while its ends will be sealed by interlocking of the grooves between the cap and the cylinder. This is done so that there is minimum leakage and to make the ERDD reusable, this patch can be replaced.
2. Core of the support: A support mechanism is designed on which the fan and needles are to be mounted. The needles will tear the mylar patch whilst the fan will suck out helium in the primary cylinder and suck in the air in the secondary cylinder. Grooves are engraved on the support as well.
3. Telescopic Rod: The support is then connected to the telescopic rod which is driven by the motor. The motor twists the rod which unscrews and opens up as the core supports unscrews and progresses forward in the cylinder.
4. Motor Casing: To hold the motor in place.
5. Revolve disc: Two discs are designed to attach the cylinder into the mylar and ensure no leakage.

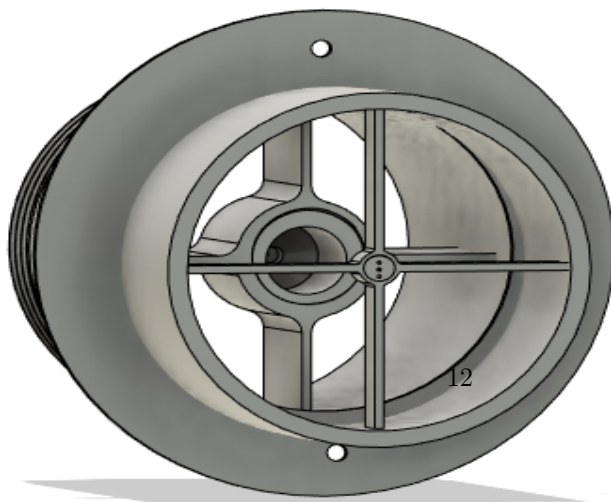


(a) Front View



(b) Side View

Figure 8: Views



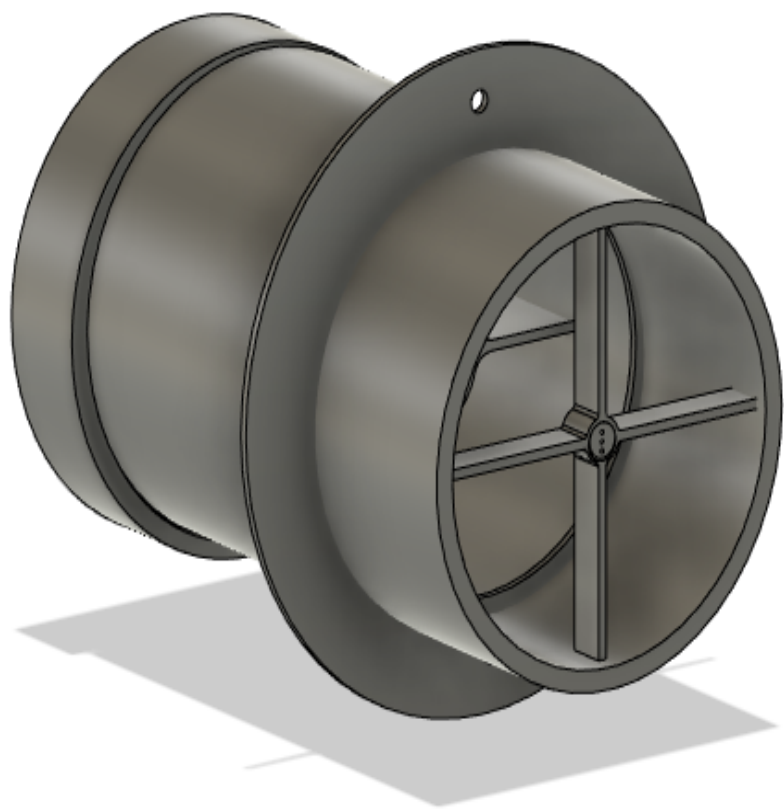


Figure 10: Isometric View-2



Figure 11: 3D printed components

2.6 Advantages

1. This system makes the ERDD reusable as specified in our mission statement. We just have to change the sheet of mylar again.
2. Due to the fan the rate at which helium will escape will be more and the aerostat will deflate at a faster rate.
3. The fan at the bottom will fill in air, making the aerostat heavier which will help to bring down the aerostat.
4. The deflation rate can be controlled by changing the RPM of motor with propellers attached.
5. This system contains lightweight electronics making the overall system lighter.
6. This system can be remote controlled and be automated at the same time.
7. The electronic system used is cost effective.

2.7 Disadvantages

1. There are chances of helium leakage in this system. While reapplying the mylar sheet there may be some human errors which may cause helium leakage
2. Once deployed, our ERDD system can not be reused midway of the deflation.